



Indian Institute of Technology Kanpur
Department of Mathematics and Statistics
Introduction to Bayesian Analysis (MTH422)
End-semester examination, Date: May 1, 2025, Thursday

Time: 8:00–10:00 AM (8:00–10:40 AM for DAP students) Spring 2025 Max point: 25

The answers need to be explicit. Writing some philosophical answer and claiming that it is correct will lead to no point. Calculation mistakes will be penalized.

1. What are aleatoric and epistemic uncertainties? Explain the difference with one concrete example. (2 points)
2. Consider a sample of size n , say, $X_1, \dots, X_n$ from the normal density given by $f(x|\sigma_1) = \sigma_1^{-1} \phi(x/\sigma_1)$, where $\phi(\cdot)$ is the standard normal density. We choose improper prior $\pi(\sigma_1^2) = \sigma_1^{-2}$ for all $\sigma_1 \in \mathbb{R}^+$. Find the posterior mean of σ_1^2 . You need to derive it explicitly. (2 points)
3. Now consider a sample of size n , say, $Y_1, \dots, Y_n$ from the mixture of two normal densities given by $f(y|\sigma_1) = 0.25\sigma_1^{-1}\phi(y/\sigma_1) + 0.75\sigma_2^{-1}\phi(y/\sigma_2)$. Here σ_2 is fixed and known. We choose improper prior $\pi(\sigma_1) = \sigma_1^{-2}$ for all $\sigma_1 \in \mathbb{R}^+$. Find the posterior mean of σ_1^2 . You need to derive it explicitly. (2 points)
4. Suppose we have observed the data $\mathcal{D} = \{(X_i, Y_i), i = 1, \dots, n\}$ where Y_i are responses and considered to be random and X_i are predictors and considered to be deterministic and varying over the range $X_i \in [a, b]$. We model the nonparametric regression model by $Y(x) = f(x) + \varepsilon(x)$, where $\varepsilon(x) \stackrel{\text{iid}}{\sim} \text{Normal}(0, \sigma^2)$. Show that a Gaussian process (GP) prior leads to conjugacy, i.e., if we consider a GP prior for $f(\cdot)$, the posterior process $f(\cdot)|\mathcal{D}$ will also be a GP. Hint: Use the definition of a GP in terms of finite-dimensional realizations, then calculate the posterior for any arbitrary m points over its support using the posterior predictive distribution, and then show that the posterior also satisfies the properties of a GP. Be concise! (5 points)
5. We want to decide between two models

$$\mathcal{M}_1 : Y \sim \text{Poisson}(\lambda_0), \quad \mathcal{M}_2 : Y|\lambda \sim \text{Poisson}(\lambda), \quad \lambda \sim \text{Exponential}(\lambda_0^{-1}).$$

Find the Bayes factor $BF_{2,1}$ in terms of λ_0 and Y . You need to report a final closed-form expression. (3 points)

6. We consider the following Bayesian hierarchical model, which is a hidden Markov model. In the data layer, we assume $Y_t = X_t + \varepsilon_t$, where $\varepsilon_t \stackrel{\text{iid}}{\sim} \text{Normal}(0, \sigma^2)$ for $t = 1, 2, \dots$. In the process layer, we assume $X_t = \rho X_{t-1} + \eta_t$, where $\eta_t \stackrel{\text{iid}}{\sim} \text{Normal}(0, (1 - \rho^2)\tau^2)$ and $X_1 \sim \text{Normal}(0, \tau^2)$. Further, in the prior layer, we assume $\sigma^2, \tau^2 \stackrel{\text{iid}}{\sim} \text{Inverse-gamma}(0.01, 0.01)$ and $\rho \sim \text{Uniform}(-1, 1)$. Explain which ones of these layers represent aleatoric and epistemic uncertainties. No point will be given if no correct explanation is provided. This question does NOT have any partial credit; either you will get 2 for a fully correct answer or 0. (2 points)
7. Suppose we observe time series data in the setup of Question 6 for the period $t = 1$ to $t = 100$. However, the data collection instrument was not functioning for the period $t = 21, \dots, 30$ and so the data $Y_{21}, \dots, Y_{30}$ are missing. Write down a Gibbs sampling scheme, with a Metropolis-Hastings step for updating ρ . The exact algorithm needs to be reported so that it can be implemented. No point will be given for partial answers. For example, just writing a posterior distribution correctly will not earn any points. Put a box around your final forms of full conditional posteriors. No need to write the skeleton of the Gibbs sampling. (4 points)

8. Suppose a Bayesian hierarchical model for annual rainfall at Kanpur is given by

$$\begin{aligned} Y_t | Z_t &\overset{\text{Indep}}{\sim} \text{Gamma}(2, \exp[\beta_0 + \beta_1 Z_t]) \\ Z_1 | \sigma^2 &\sim \text{Normal}(0, \sigma^2), \quad Z_{t+1} | Z_t, \sigma^2, \rho \sim \text{Normal}(\rho Z_t, (1 - \rho^2)\sigma^2), t \geq 1 \\ \beta_0, \beta_1, \sigma^2, \rho &\sim \pi(\beta_0, \beta_1, \sigma^2, \rho). \end{aligned}$$

Write down a directed acyclic graph (DAG) representation of the above model. (2 points)

9. The Weibull distribution has a density function $f(x) = \alpha\beta x^{\alpha-1} \exp[-\beta x^\alpha]$ for $x \in \mathbb{R}^+$. Here, the shape and rate parameters are given by $\alpha > 0$ and $\beta > 0$. The mean of a $\text{Weibull}(\alpha_0, \beta)$ distributed random variable is given by $\Gamma(1 + \alpha_0^{-1})\beta^{-1}$ for some known $\alpha = \alpha_0$. Hence, we are interested in summarizing the posterior of β through a single measure that minimizes the posterior risk under the loss function $\ell(\hat{\beta}, \beta) = |\hat{\beta}^{-1} - \beta^{-1}|$. Find the Bayes rule. (3 points)